

SLASH MARK IT SOLUTIONS

Business Analyst Internship Project Report

Employee Attrition & Performance Analysis Using Artificial Neural Network (ANN)

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Submitted To

Slash Mark IT Solutions

Business Analyst Internship Program

Internship Details

Intern ID: SMI85689

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CERTIFICATE

This is to certify that Ms. Kavya P, a Bachelor of Engineering student in Computer Science and Business Systems (CSBS) from St. Joseph Engineering College, Mangaluru, has successfully completed the Business Analyst Internship project titled “Employee Attrition & Performance Analysis Using Artificial Neural Network (ANN)” at Slash Mark IT Solutions.

The project focuses on analyzing employee attrition using the IBM HR Analytics dataset to identify the key factors influencing employee turnover. The analysis involved data preprocessing, exploratory data analysis (EDA), feature analysis, and the development of an Artificial Neural Network (ANN) model to predict employee attrition. The findings provide valuable business insights and recommendations to support data-driven human resource decision-making.

This report is submitted in partial fulfilment of the requirements for the successful completion of the Business Analyst Internship conducted by Slash Mark IT Solutions.

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ACKNOWLEDGEMENT

I appreciate Slash Mark IT Solutions for providing the project resources and platform that enabled me to work on this internship. Working on this project helped me gain practical exposure to Exploratory Data Analysis (EDA), data preprocessing, machine learning concepts, and the role of data-driven decision-making in Human Resource Analytics.

I am also grateful to my family and friends for their continuous encouragement and motivation throughout the successful completion of this project.

Completing this internship has strengthened my analytical thinking, problem-solving abilities, and understanding of how business analytics can be applied to solve organizational challenges.

EXECUTIVE SUMMARY

Employee attrition is one of the most significant challenges faced by organizations, as it affects workforce stability, operational efficiency, recruitment costs, and overall business performance. Understanding the factors that influence employee turnover enables organizations to develop effective retention strategies and improve employee satisfaction.

This project focuses on analyzing employee attrition using the IBM HR Analytics Employee Attrition & Performance dataset. The primary objective is to identify the key factors contributing to employee attrition and develop a predictive model capable of identifying employees who are likely to leave the organization. The project combines business analysis with data analytics techniques to transform employee data into meaningful business insights.

The analysis began with data preprocessing, including validation of data quality and exploration of numerical and categorical features. Exploratory Data Analysis (EDA) was performed to understand employee demographics, job characteristics, compensation, satisfaction levels, and work-related factors. Correlation analysis and feature evaluation were then conducted to identify variables that significantly influence employee attrition.

An Artificial Neural Network (ANN) model was developed to predict employee attrition based on employee attributes. The model achieved an overall prediction accuracy of 88.45%, demonstrating its effectiveness in classifying employees who are likely to remain with the organization or leave.

The analysis indicates that factors such as monthly income, job level, work-life balance, job involvement, environment satisfaction, relationship satisfaction, and overtime have a significant influence on employee retention. Based on these findings, practical recommendations have been proposed to support Human Resource departments in improving employee engagement, enhancing workplace satisfaction, and implementing proactive retention strategies.

Overall, this project demonstrates how Business Analytics and Machine Learning can support evidence-based decision-making in Human Resource Management by enabling organizations to identify attrition risks and take preventive actions before employee turnover occurs.

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CHAPTER 1: INTRODUCTION

1.1 Introduction

In today's competitive business environment, employees are considered one of the most valuable assets of an organization. The success of any organization depends not only on its financial resources and technological capabilities but also on its ability to attract, develop, and retain talented employees. A stable workforce contributes to higher productivity, improved organizational performance, and sustainable business growth.

Employee attrition refers to the reduction in an organization's workforce when employees voluntarily or involuntarily leave their jobs. While a certain level of attrition is expected in every organization, a consistently high attrition rate can result in increased recruitment costs, training expenses, loss of experienced employees, reduced productivity, and disruption of business operations. Therefore, understanding the factors that contribute to employee attrition has become an important objective for Human Resource (HR) departments.

The rapid growth of data analytics and machine learning has transformed traditional HR practices into HR Analytics, enabling organizations to make data-driven decisions. By analyzing employee-related data, organizations can identify patterns associated with employee turnover, predict potential attrition, and implement proactive retention strategies before employees decide to leave.

This project focuses on analyzing employee attrition using the IBM HR Analytics Employee Attrition & Performance dataset. The analysis combines Business Analytics techniques with Machine Learning to examine various employee attributes, including demographic information, job characteristics, compensation, job satisfaction, work-life balance, and performance-related factors. An Artificial Neural Network (ANN) model is developed to predict employee attrition based on these attributes.

The insights generated through this analysis can assist Human Resource professionals in identifying employees at higher risk of attrition, understanding the underlying causes of employee turnover, and implementing effective retention strategies that contribute to improved organizational performance.

1.2 Background

Employee attrition is a major concern across industries because replacing skilled employees requires significant investment in recruitment, onboarding, training, and knowledge transfer. Organizations are increasingly adopting predictive analytics to identify employees who are most likely to leave and to develop targeted initiatives that improve employee engagement and retention.

Human Resource Analytics combines statistical analysis, visualization techniques, and predictive modeling to transform employee data into actionable business insights. By identifying patterns in employee behavior and workplace conditions, organizations can make informed decisions that enhance workforce planning and long-term business success.

CHAPTER 2: PROBLEM STATEMENT, OBJECTIVES AND SCOPE

2.1 Problem Statement

Employee attrition is one of the major challenges faced by organizations across various industries. A high attrition rate leads to increased recruitment and training costs, loss of experienced employees, reduced productivity, and disruption of business operations. Identifying employees who are likely to leave and understanding the factors contributing to their decision are critical for improving employee retention and organizational performance.

Traditional methods of identifying employee attrition often rely on manual analysis and historical observations, making it difficult for Human Resource (HR) departments to take timely preventive actions. With the increasing availability of employee data, organizations can leverage Business Analytics and Machine Learning techniques to identify hidden patterns, predict employee attrition, and support data-driven decision-making.

This project addresses the business problem by analyzing employee data to identify the key factors influencing attrition and by developing a predictive model that can assist HR professionals in recognizing employees who may be at risk of leaving the organization.

2.2 Objectives

The primary objectives of this project are:

- To analyze employee data and understand the factors influencing employee attrition.
- To perform data preprocessing and exploratory data analysis (EDA) on the IBM HR Analytics dataset.
- To identify patterns and relationships between employee characteristics and attrition.
- To evaluate the impact of demographic, organizational, and job-related factors on employee retention.
- To develop an Artificial Neural Network (ANN) model for predicting employee attrition.
- To evaluate the performance of the predictive model using appropriate evaluation metrics.
- To provide business recommendations that support employee retention and improve organizational decision-making.

2.3 Scope of the Project

The scope of this project is limited to the analysis of the IBM HR Analytics Employee Attrition dataset. The project focuses on identifying significant factors associated with employee attrition using exploratory data analysis and predictive modeling techniques.

The analysis includes employee demographic information, job-related attributes, income, satisfaction levels, work-life balance, experience, and organizational factors. Based on these variables, an Artificial Neural Network model is developed to predict whether an employee is likely to leave the organization.

The findings of this project can assist Human Resource departments in designing effective employee retention strategies. However, the analysis is based solely on the available dataset and does not

include external factors such as economic conditions, organizational policy changes, or industry-specific trends.

2.4 Business Significance

Employee attrition directly affects organizational performance by increasing operational costs and reducing workforce stability. By identifying employees who are at higher risk of leaving, organizations can implement proactive retention strategies, improve employee engagement, and optimize talent management.

The application of Business Analytics in Human Resource Management enables organizations to move from reactive decision-making to predictive decision-making. The insights generated from this project can help HR professionals prioritize employee retention initiatives, improve workplace satisfaction, and support long-term organizational growth.

CHAPTER 3: DATASET DESCRIPTION AND PROJECT METHODOLOGY

3.1 Dataset Description

The analysis in this project is based on the IBM HR Analytics Employee Attrition & Performance dataset, which is a publicly available dataset widely used for Human Resource Analytics and machine learning applications. The dataset contains employee demographic information, job-related details, compensation, performance metrics, and satisfaction levels that help analyze the factors contributing to employee attrition.

The primary objective of using this dataset is to identify employees who are likely to leave the organization and understand the business factors influencing employee turnover. The dataset provides valuable insights into employee behavior and supports the development of predictive models for employee retention.

Dataset Summary

Attribute	Description
Dataset Name	IBM HR Analytics Employee Attrition & Performance
Source	IBM HR Analytics Dataset
Total Records	1,470 Employees
Total Features	35 Attributes
Target Variable	Attrition (Yes / No)
Data Type	Numerical and Categorical
Missing Values	No Missing Values

3.2 Important Features Used in the Analysis

The dataset contains several employee-related attributes. Some of the important variables analyzed in this project include:

- Age, Attrition, Department, Job Role, Gender
- Monthly Income, Job Level, Business Travel, Overtime
- Total Working Years, Years at Company, Years in Current Role, Years Since Last Promotion
- Work-Life Balance, Job Satisfaction, Job Involvement
- Environment Satisfaction, Relationship Satisfaction, Training Times Last Year

These attributes provide valuable information for understanding employee characteristics and identifying factors associated with employee attrition.

3.3 Project Methodology

The project was carried out using a structured Business Analytics and Machine Learning workflow to identify the factors influencing employee attrition and develop a predictive model. The methodology followed is illustrated in Figure 3.1 and described in the steps below.

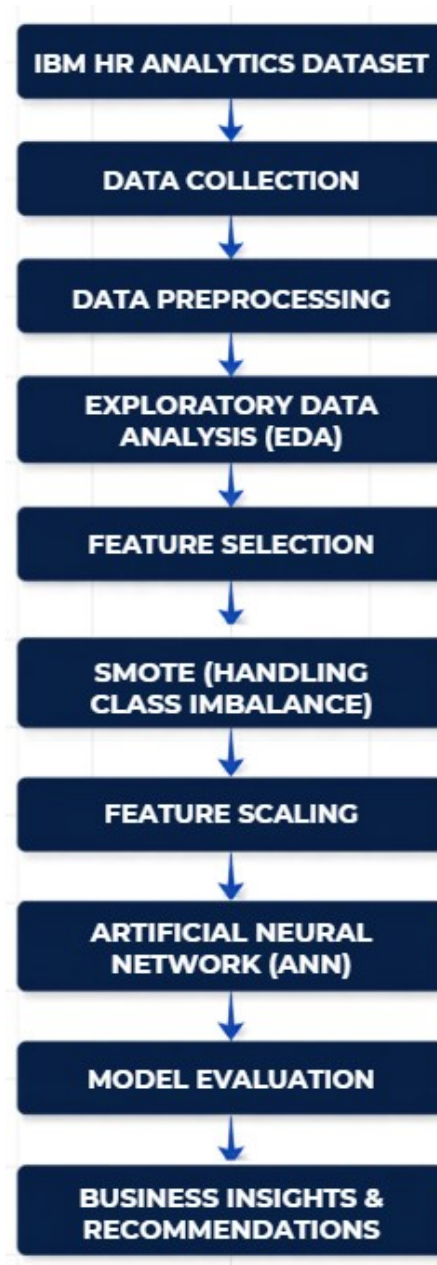


Figure 3.1: Project Methodology Workflow

Step 1: Data Collection

The IBM HR Analytics Employee Attrition & Performance dataset was collected from Kaggle. The dataset contains employee demographic information, job-related attributes, compensation details, performance indicators, and satisfaction-related variables.

Step 2: Data Preprocessing

The dataset was inspected to understand its structure and quality. Missing values were checked, numerical features were examined using box plots to identify outliers, and the dataset was prepared for further analysis.

Step 3: Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand employee demographics, compensation patterns, work-life balance, satisfaction levels, and other organizational characteristics. Various visualizations were used to identify relationships between employee attributes and attrition.

Step 4: Feature Engineering

Relevant features contributing to employee attrition were selected. Categorical variables were converted into numerical representations, and the dataset was prepared for machine learning.

Step 5: Handling Class Imbalance

Since the dataset contained significantly fewer employees who left the organization than employees who stayed, the Synthetic Minority Oversampling Technique (SMOTE) was applied to balance the classes.

Step 6: Feature Scaling

Numerical variables were standardized using feature scaling to improve the performance and convergence of the Artificial Neural Network model.

Step 7: Model Development

An Artificial Neural Network (ANN) model was developed to classify employees into two categories: employees likely to remain with the organization and employees likely to leave.

Step 8: Model Evaluation

The trained ANN model was evaluated using appropriate performance metrics. The overall prediction accuracy obtained was 88.45%.

Step 9: Business Insights and Recommendations

The analytical findings were interpreted from a business perspective to identify major factors influencing employee attrition and to propose practical recommendations for improving employee retention.

3.4 Software and Tools Used

Tool / Technology	Purpose
Python	Data Analysis and Machine Learning
Pandas	Data Manipulation
NumPy	Numerical Computation
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-learn	Data Preprocessing and Feature Selection
TensorFlow / Keras	Artificial Neural Network Model
Jupyter Notebook / VS Code	Model Development

Tool / Technology	Purpose
GitHub	Project Repository

CHAPTER 4: DATA PREPROCESSING

4.1 Introduction

Data preprocessing is an essential step in every data analytics and machine learning project. It ensures that the dataset is accurate, consistent, and suitable for analysis by identifying missing values, detecting outliers, and understanding the structure of the data before applying analytical techniques.

In this project, the IBM HR Analytics Employee Attrition dataset was examined to verify data quality. The preprocessing stage included dataset inspection, missing value analysis, and outlier detection. Since the dataset was well structured and contained no missing values, only minimal preprocessing was required before proceeding to Exploratory Data Analysis (EDA).

4.2 Dataset Inspection

The dataset was initially loaded into the notebook environment using the Pandas library. Basic inspection techniques were performed to understand the structure and characteristics of the dataset. The following operations were carried out:

- Importing the required Python libraries.
- Loading the IBM HR Analytics dataset.
- Displaying the first few records of the dataset.
- Examining the dimensions of the dataset.
- Identifying the data types of each feature.
- Verifying the presence of missing values.

These initial steps ensured that the dataset was correctly loaded and ready for further preprocessing.

4.3 Missing Value Analysis

One of the primary preprocessing tasks was to examine the dataset for missing or null values.

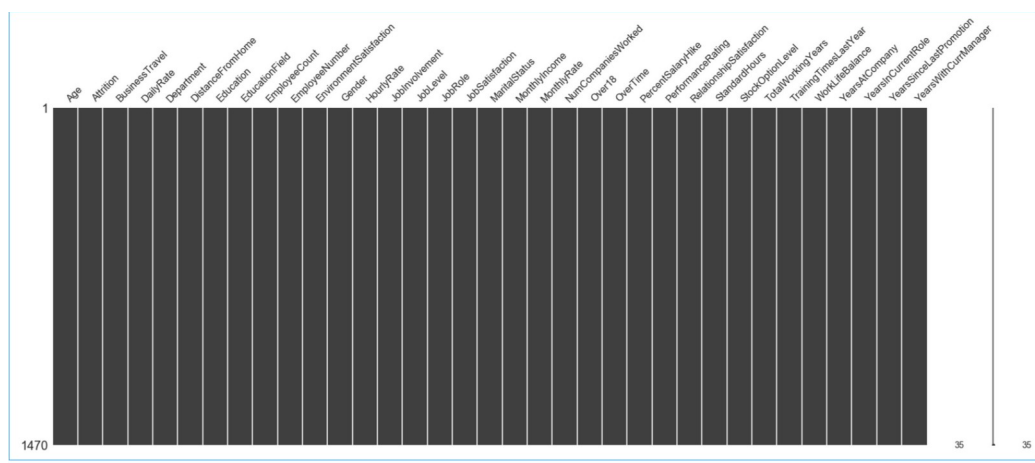


Figure 4.1: Missing Values Matrix

Observation

The matrix plot generated for the dataset shows a uniformly solid, unbroken pattern across all 35 columns, spanning the full range of 1 to 1,470 records. There are no visible white streaks or gaps within any column, which is the visual signature that a missingness matrix uses to flag incomplete entries.

Inference

Because every column renders as a continuous block, the dataset can be confirmed as fully populated. This removes the need for imputation, row deletion, or any other missing-data handling technique before moving into exploratory analysis — a convenient starting point that allowed the analysis to proceed directly to distributional checks.

4.4 Outlier Detection

To understand the distribution of numerical variables and identify possible outliers, box plots were generated for all numerical features on a single combined chart.

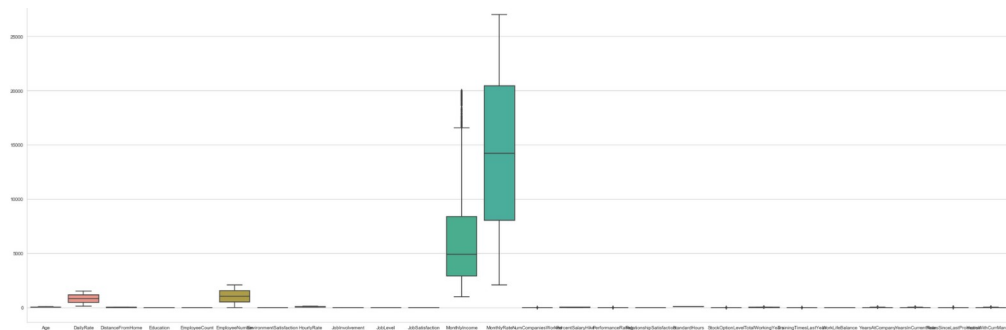


Figure 4.2: Distribution of Numerical Features Using Box Plots

Observation

With all 35 numerical columns plotted on a single chart, four features are distinguishable as visible boxes while the rest compress into thin lines near zero. DailyRate forms a modest box roughly between 200 and 1,200. EmployeeNumber shows a similarly sized box in the 400–2,000 range. MonthlyIncome stands out with a box spanning approximately 3,000 to 8,500 and a whisker extending to about 20,000. MonthlyRate is the largest of all, with a box from roughly 8,000 to 20,500 and a whisker reaching close to 27,000. Variables such as Age, Education, JobLevel, and the various satisfaction ratings are compressed to near-invisible lines at the bottom of the scale.

Inference

The chart confirms that the dataset's numerical variables operate on very different scales — income-related fields alone span tens of thousands, while satisfaction and rating fields are bounded within single digits. A combined box plot is therefore not an effective way to compare distributions directly; this observation justified plotting each numerical variable individually during exploratory data analysis and later applying feature scaling before model training.

4.5 Summary of Data Preprocessing

The preprocessing stage confirmed that the dataset was well-structured and free from missing values. Initial visualizations provided insights into the distribution of numerical features and highlighted

potential outliers for further consideration during model development. These preprocessing steps established a reliable foundation for exploratory data analysis and predictive modeling.

Chapter Summary

The data preprocessing phase ensured that the IBM HR Analytics dataset was clean, consistent, and suitable for analysis. Since no missing values were detected, no additional data imputation techniques were required. Furthermore, the distribution of numerical features was examined using box plots to gain an initial understanding of the dataset before proceeding to Exploratory Data Analysis (EDA).

CHAPTER 5: EXPLORATORY DATA ANALYSIS (EDA)

5.1 Introduction

Exploratory Data Analysis (EDA) is a fundamental step in the data analysis process that involves examining the dataset using statistical summaries and visualizations to identify patterns, trends, relationships, and anomalies. It enables analysts to gain a deeper understanding of the data before applying predictive models.

In this project, EDA was performed to examine the characteristics of employee demographics, job-related factors, compensation, satisfaction levels, and organizational attributes. The analysis helps identify potential factors associated with employee attrition and provides valuable business insights that support data-driven Human Resource decisions.

5.2 Analysis of Numerical Features

The distribution of important numerical variables was analyzed to understand employee demographics, work experience, income distribution, and career progression.

5.2.1 Age Distribution

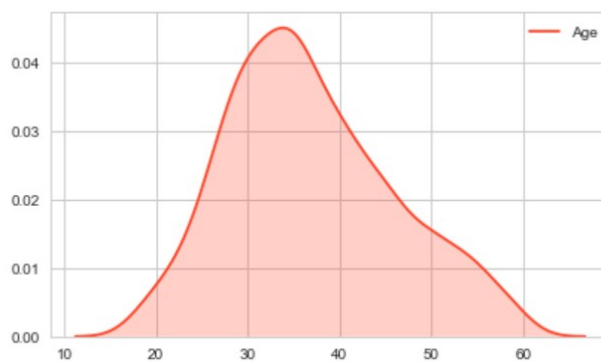


Figure 5.1: Distribution of Age Distribution

Observation

The Age feature was analyzed using Kernel Density Estimation (KDE). The resulting curve is unimodal and right-skewed, rising steadily from around 18 years, peaking sharply near 34–35, and tapering off gradually through the 50s before flattening out close to 60. The bulk of the density sits between roughly 28 and 42 years.

Business Interpretation

The organization consists primarily of early-to-mid career professionals rather than a workforce evenly spread across age groups. Understanding this concentration helps HR departments evaluate workforce demographics and later analyze whether specific age groups are more likely to leave the organization.

5.2.2 Total Working Years

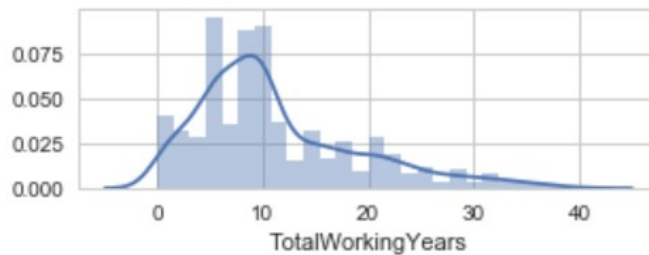


Figure 5.2: Distribution of Total Working Years

Observation

The distribution is heavily right-skewed, with a sharp peak between 5 and 10 years of total experience and a long tail extending out toward 40 years. Very few employees fall beyond the 30-year mark.

Business Interpretation

Most employees are past the early-career stage but have not yet reached long-tenure status. This experienced-but-not-veteran profile is a segment organizations often need to actively retain, since such employees are typically the most marketable to competitors.

5.2.3 Monthly Income Distribution

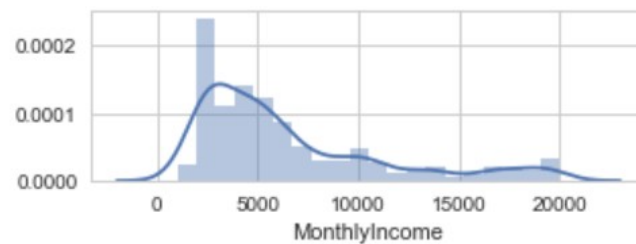


Figure 5.3: Distribution of Monthly Income Distribution

Observation

Income is strongly right-skewed, with a tall concentration below roughly 5,000 and a much flatter, thinner spread of employees earning between 10,000 and 20,000. A small secondary bump appears near the 19,000–20,000 range.

Business Interpretation

The compensation structure is bottom-heavy, with a large proportion of the workforce in lower income bands. This distribution makes income a variable worth watching closely for its relationship with attrition risk, which is examined further in Section 5.4.

5.2.4 Years at Company

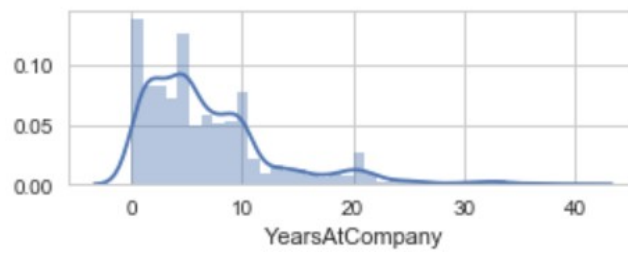


Figure 5.4: Distribution of Years at Company

Observation

The distribution peaks sharply around 2–5 years and declines steeply afterward, with a small secondary spike near the 20-year mark and a long, thin tail beyond that.

Business Interpretation

Organizational tenure is generally short, suggesting either a relatively young company or elevated turnover in the early years of employment. The small cluster of long-tenured employees around 20 years may represent a distinct, highly loyal segment worth studying separately from the broader workforce.

5.2.5 Distance from Home

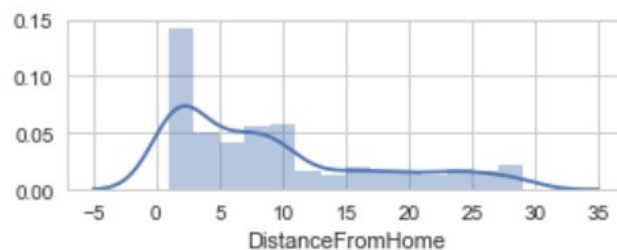


Figure 5.5: Distribution of Distance from Home

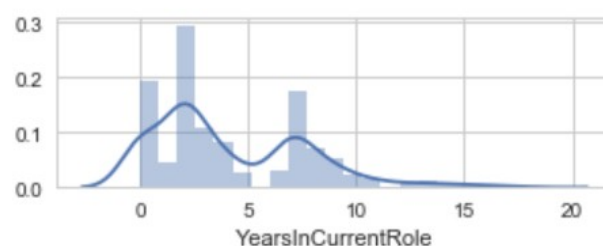
Observation

The distribution peaks at a short commuting distance (around 1–2 units) and declines from there, though it does not taper off cleanly — a secondary, flatter bump is visible between roughly 8 and 12, and a smaller one again near 25–29.

Business Interpretation

While most employees live close to the workplace, there are identifiable pockets of employees with considerably longer commutes. These longer-distance groups are worth cross-referencing against attrition, since commute burden is a commonly cited driver of job dissatisfaction.

5.2.6 Years in Current Role

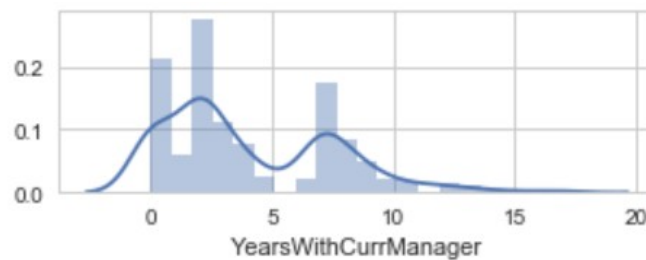


*Figure 5.6: Distribution of Years in Current Role***Observation**

The distribution is distinctly bimodal — a tall peak around 2 years and a smaller second peak around 7 years, with a dip in between near 4–5 years.

Business Interpretation

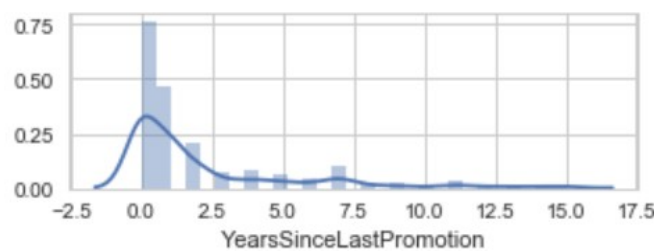
The two peaks likely correspond to different career-stage groups: employees newly placed into a role, and a smaller group who have settled into their position for an extended period without a role change. HR could examine whether the dip between the two peaks reflects a point where employees are promoted, transferred, or lost to attrition.

5.2.7 Years with Current Manager*Figure 5.7: Distribution of Years with Current Manager***Observation**

Similar to years in current role, this distribution also shows two peaks — one near 2 years and a smaller one around 7 years — with the majority of employees falling in the shorter range.

Business Interpretation

Short manager tenure is common across the workforce, which may reflect internal restructuring or reporting-line changes. Since managerial relationships are frequently linked to engagement, this variable is a reasonable candidate for further attrition analysis.

5.2.8 Years Since Last Promotion*Figure 5.8: Distribution of Years Since Last Promotion***Observation**

The distribution is extremely right-skewed, with a very tall spike at or near 0 years and a long, low tail extending out to nearly 15–17 years. Small secondary bumps appear around 1–2 and 7 years.

Business Interpretation

A large share of the workforce was promoted recently, but a visible minority has gone many years without advancement. This long tail represents a potential at-risk group — employees who may feel their career progression has stalled.

5.2.9 Percent Salary Hike



Figure 5.9: Distribution of Percent Salary Hike

Observation

The distribution peaks between roughly 11% and 14%, then declines steadily with a long, gently sloped tail extending out to about 25%.

Business Interpretation

Salary increments cluster tightly around a standard, modest range, with larger hikes being comparatively rare. This pattern suggests a fairly standardized compensation review process rather than highly individualized raises, which is useful context when interpreting income's relationship with attrition.

5.2.10 Training Times Last Year

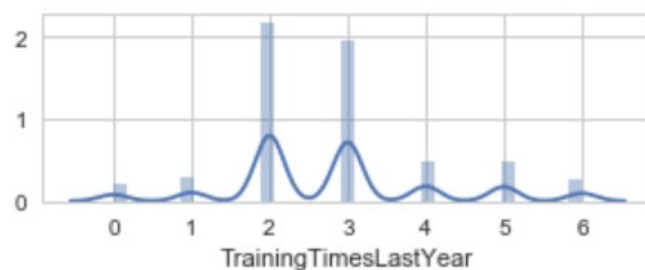


Figure 5.10: Distribution of Training Times Last Year

Observation

This distribution is distinctly multi-modal with sharp spikes at specific integer values — the tallest spikes occur at 2 and 3 training sessions, with smaller peaks at 0, 4, and 5, and a noticeably shorter one at 6.

Business Interpretation

Training exposure is not evenly distributed — most employees attend 2 to 3 sessions annually, while a smaller group receives either minimal or above-average training. Because professional development is often linked to engagement, this unevenness is worth examining alongside attrition outcomes.

Chapter Summary (Numerical Features)

The numerical feature analysis provided a comprehensive understanding of employee demographics, experience, compensation, and career progression. These insights establish a strong foundation for analyzing employee attrition by highlighting workforce characteristics and identifying variables that may influence employee retention.

5.3 Analysis of Categorical Features

Categorical features provide valuable insights into employee characteristics, organizational structure, and workplace conditions. These variables help identify patterns associated with employee attrition by comparing employee groups across different categories. The analysis of categorical features enables Human Resource professionals to understand workforce composition and identify potential risk factors influencing employee retention.

5.3.1 Attrition

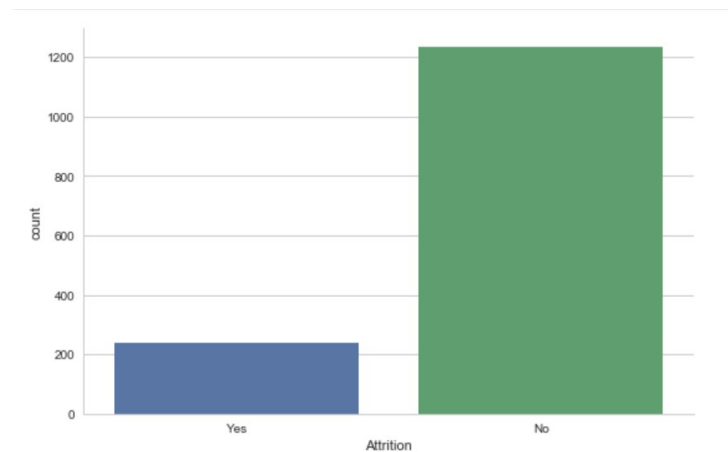


Figure 5.11: Distribution of Employee Attrition

Observation

The count plot shows approximately 1,233 employees under “No” against roughly 237 under “Yes” — a ratio of more than 5 to 1 in favor of employees who stayed.

Business Interpretation

This confirms a pronounced class imbalance in the target variable. Left untreated, a predictive model trained on this data would be biased toward predicting “No” for nearly every employee. This is precisely why the Synthetic Minority Oversampling Technique (SMOTE) was applied later in the preprocessing pipeline (Section 6.3) to rebalance the minority class before model training.

5.3.2 Business Travel Frequency

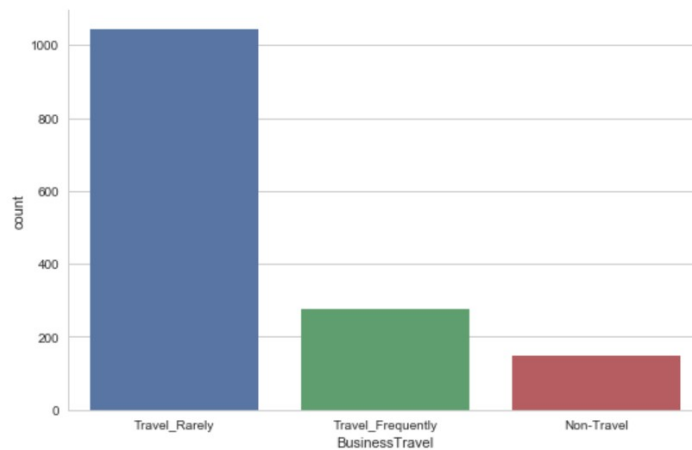


Figure 5.12: Distribution of Employee Business Travel Frequency

Observation

“Travel_Rarely” dominates the chart at just over 1,000 employees, “Travel_Frequently” accounts for roughly 275, and “Non-Travel” is the smallest group at around 150.

Business Interpretation

The organization's travel policy affects only a minority of the workforce heavily. Even so, the “Travel_Frequently” segment — while smaller — may carry disproportionate attrition risk given the added strain frequent travel places on work-life balance.

5.3.3 Overtime

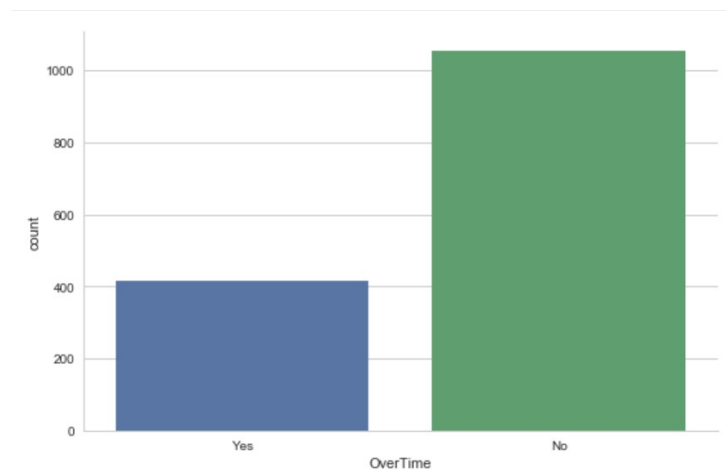


Figure 5.13: Distribution of Employee Overtime

Observation

Roughly 1,054 employees report not working overtime, compared to about 416 who do — meaning close to a quarter of the workforce is regularly working beyond standard hours.

Business Interpretation

While the overtime group is the minority, it still represents a substantial share of the organization. Given the well-documented link between overtime and burnout, this segment deserves proportionally closer attention when assessing attrition drivers.

5.3.4 Department

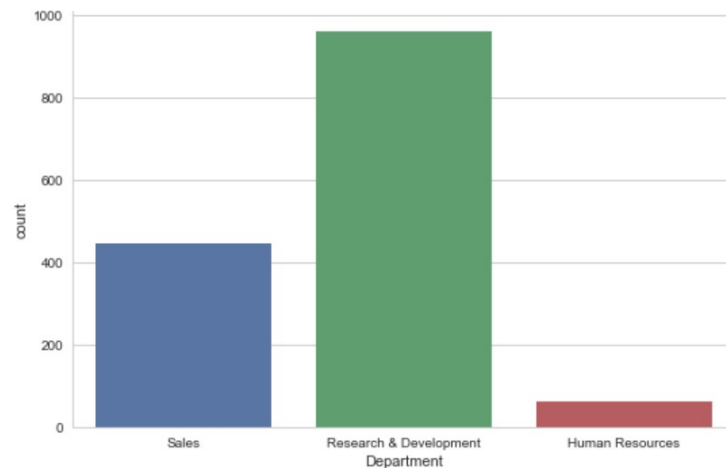


Figure 5.14: Distribution of Employee Department

Observation

Research & Development is by far the largest department, with just under 1,000 employees, followed by Sales at roughly 450. Human Resources is a small fraction of the workforce, with a headcount close to 60.

Business Interpretation

The organization's employee base is concentrated in R&D functions, meaning attrition trends in this department will disproportionately shape overall company-wide attrition figures. Retention strategy should be weighted accordingly, since losing even a small percentage of R&D staff represents a larger absolute headcount loss than the same percentage in HR.

5.3.5 Educational Background

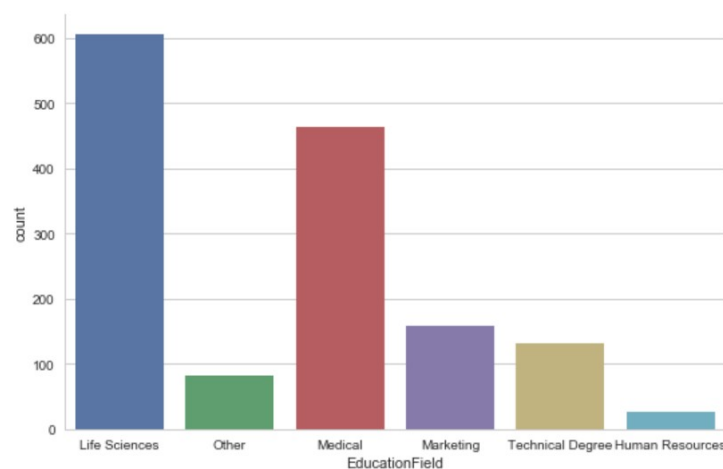


Figure 5.15: Distribution of Employee Educational Background

Observation

Life Sciences is the most common educational background (just over 600 employees), followed closely by Medical (around 465). Marketing and Technical Degree together account for a moderate share, while Other and Human Resources fields are comparatively small.

Business Interpretation

The workforce's academic composition mirrors a life-sciences and medical-oriented business, consistent with the department distribution above. This context is useful when interpreting job-role attrition, since specific educational backgrounds tend to map onto specific roles.

5.3.6 Gender

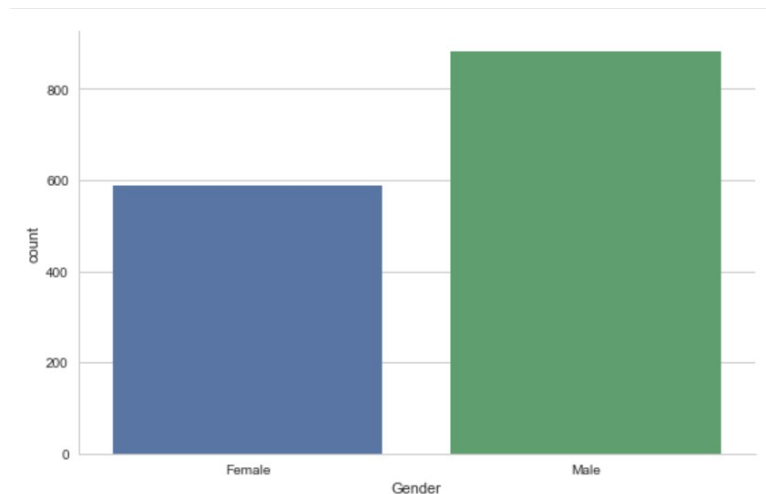


Figure 5.16: Distribution of Employee Gender

Observation

Male employees number just under 900, compared to female employees at roughly 590 — a split of approximately 60:40.

Business Interpretation

The gender split is moderately imbalanced but not extreme. As explored in Section 5.4.1, this imbalance in headcount does not translate into a meaningfully different attrition rate between the two groups.

5.3.7 Job Role

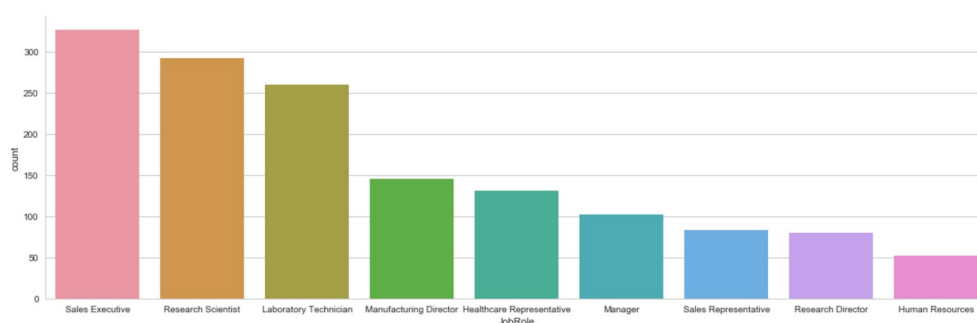


Figure 5.17: Distribution of Employee Job Role

Observation

Sales Executive is the largest single role (approximately 325 employees), followed by Research Scientist (~293) and Laboratory Technician (~260). Roles such as Research Director and Human Resources sit at the bottom of the distribution, each under 100 employees.

Business Interpretation

Three roles — Sales Executive, Research Scientist, and Laboratory Technician — together account for a large share of total headcount. Retention initiatives targeted at these three roles would have the broadest organizational impact, simply due to their scale.

5.3.8 Summary of Categorical Feature Analysis

The categorical feature analysis provides an overview of workforce composition across departments, job roles, educational backgrounds, gender, business travel frequency, overtime, and employee attrition. These insights establish a foundation for understanding employee characteristics before examining the relationship between these variables and employee attrition.

5.4 Relationship Between Employee Attributes and Attrition

The previous sections described employee characteristics independently. In this section, categorical variables are analyzed together with employee attrition to identify which organizational and demographic factors are associated with employees leaving the organization.

5.4.1 Gender vs Employee Attrition

Attrition	No	Yes
Gender		
Female	0.852041	0.147959
Male	0.829932	0.170068
All	0.838776	0.161224

Figure 5.18: Gender vs Employee Attrition

Observation

Female employees show an attrition rate of 14.80%, compared to 17.01% for male employees, against an overall baseline attrition rate of 16.12%.

Business Interpretation

The gap between the two groups is under three percentage points, which is not a substantial difference. This confirms gender is not a strong standalone predictor of attrition and should be interpreted only in combination with other variables, not as an isolated retention lever.

5.4.2 Job Level vs Employee Attrition

Attrition	No	Yes
JobLevel		
1	0.736648	0.263352
2	0.902622	0.097378
3	0.853211	0.146789
4	0.952830	0.047170
5	0.927536	0.072464
All	0.838776	0.161224

Figure 5.19: Job Level vs Employee Attrition

Observation

Attrition is highest at Job Level 1, at 26.34%, and falls to 9.74% at Level 2, 14.68% at Level 3, and reaches its lowest point at Level 4, at 4.72%, before rising slightly again to 7.25% at Level 5.

Business Interpretation

The steep drop between Level 1 and every level above it is the standout pattern here — entry-level employees leave at roughly three to five times the rate of employees at Levels 2 through 5. This points to the first one to two years of employment as the highest-risk window for attrition, making early-career retention programs a high-leverage intervention.

5.4.3 Environment Satisfaction vs Attrition

Attrition	No	Yes
EnvironmentSatisfaction		
1	0.746479	0.253521
2	0.850174	0.149826
3	0.863135	0.136865
4	0.865471	0.134529
All	0.838776	0.161224

Figure 5.20: Environment Satisfaction vs Attrition

Observation

Attrition decreases consistently as environment satisfaction rises: 25.35% at the lowest satisfaction rating (1), down to 14.98% at rating 2, 13.69% at rating 3, and 13.45% at rating 4.

Business Interpretation

The relationship is clearly monotonic — every step up in reported environment satisfaction corresponds to a drop in attrition. The steepest drop occurs between ratings 1 and 2, suggesting that even a modest improvement for the most dissatisfied employees would yield an outsized retention benefit.

5.4.4 Job Involvement vs Attrition

	Attrition	No	Yes
JobInvolvement			
1	0.662651	0.337349	
2	0.810667	0.189333	
3	0.855991	0.144009	
4	0.909722	0.090278	
All	0.838776	0.161224	

Figure 5.21: Job Involvement vs Attrition

Observation

Employees with the lowest job involvement rating (1) show a 33.73% attrition rate — the highest of any single sub-group examined in this analysis — falling to 18.93% at rating 2, 14.40% at rating 3, and 9.03% at rating 4.

Business Interpretation

Job involvement shows the strongest gradient of all the satisfaction-type variables analyzed: attrition among the least-involved employees is nearly four times that of the most-involved group. This makes job involvement one of the more actionable levers available to HR, since involvement can be influenced directly through task ownership, recognition, and participation in decision-making.

5.4.5 Work-Life Balance vs Attrition

	Attrition	No	Yes
WorkLifeBalance			
1	0.687500	0.312500	
2	0.831395	0.168605	
3	0.857783	0.142217	
4	0.823529	0.176471	
All	0.838776	0.161224	

Figure 5.22: Work-Life Balance vs Attrition

Observation

Attrition is highest at the lowest work-life balance rating (1), at 31.25%. It drops sharply to 16.86% at rating 2 and 14.22% at rating 3, but rises again slightly to 17.65% at rating 4.

Business Interpretation

Unlike environment satisfaction and job involvement, this relationship is not strictly linear — the lowest attrition occurs at a moderate rating (3) rather than the maximum work-life balance score. The dominant finding, however, remains that employees reporting poor work-life balance (rating 1) leave at close to double the rate of every other group.

5.4.6 Relationship Satisfaction vs Attrition

	Attrition	No	Yes
RelationshipSatisfaction			
1	0.793478	0.206522	
2	0.851485	0.148515	
3	0.845316	0.154684	
4	0.851852	0.148148	
All	0.838776	0.161224	

Figure 5.23: Relationship Satisfaction vs Attrition

Observation

Attrition is highest at the lowest relationship satisfaction rating (1), at 20.65%, and settles into a narrower band of 14.81%–15.47% across ratings 2 through 4.

Business Interpretation

The main effect here is concentrated at the bottom of the scale — employees reporting the poorest workplace relationships leave at a noticeably higher rate, while satisfaction levels 2 and above show comparatively little differentiation between each other. This suggests interventions should prioritize identifying and supporting the specific employees at the lowest relationship-satisfaction tier, rather than uniform relationship-building initiatives across the board.

Chapter Summary

The exploratory analysis of categorical variables identified several organizational factors associated with employee attrition. Variables such as job level, work-life balance, job involvement, environment satisfaction, relationship satisfaction, and overtime demonstrate meaningful relationships with employee retention. These findings provide valuable business insights that guide the subsequent predictive modeling stage and support evidence-based Human Resource decision-making.

CHAPTER 6: FEATURE ENGINEERING AND PREDICTIVE MODELING

6.1 Introduction

After completing the Exploratory Data Analysis (EDA), the next stage involved preparing the dataset for predictive modeling. Raw employee data cannot be directly used for machine learning because categorical variables must be converted into numerical values, class imbalance must be addressed, and the data should be transformed into a suitable format for model training.

This chapter describes the preprocessing techniques, feature engineering methods, and Artificial Neural Network (ANN) model used to predict employee attrition.

6.2 Feature Selection

Feature selection involves identifying the variables that contribute most significantly to predicting employee attrition. Selecting relevant features improves model performance by reducing unnecessary information and focusing on the most influential employee characteristics.

Based on the exploratory analysis conducted in Chapter 5, variables such as Monthly Income, Job Level, Age, Overtime, Work-Life Balance, Job Involvement, Environment Satisfaction, Relationship Satisfaction, Years at Company, and Total Working Years demonstrated meaningful relationships with employee attrition. These variables formed the primary inputs for developing the Artificial Neural Network model.

6.3 Handling Class Imbalance Using SMOTE

The employee attrition dataset is highly imbalanced, with substantially more employees remaining in the organization than employees leaving, as shown in Section 5.3.1. Training a machine learning model directly on such data may bias predictions toward the majority class.

To overcome this issue, the Synthetic Minority Oversampling Technique (SMOTE) was applied. SMOTE generates synthetic samples of the minority class, resulting in a more balanced dataset and enabling the model to learn attrition patterns more effectively.

Business Importance

Applying SMOTE improves the model's ability to correctly identify employees who are likely to leave the organization rather than simply predicting the majority class.

6.4 Feature Scaling

The numerical features in the dataset vary considerably in scale. Variables such as Monthly Income have much larger numerical values than satisfaction ratings or job levels. Without scaling, variables with larger values may dominate the learning process of the Artificial Neural Network.

Therefore, feature scaling was performed using StandardScaler, which standardizes numerical variables to have a mean of zero and a standard deviation of one. Standardization improves model convergence and contributes to better prediction performance.

6.5 Artificial Neural Network (ANN) Model

An Artificial Neural Network (ANN) was developed to classify employees into two categories:

- Employees likely to remain with the organization
- Employees likely to leave the organization

The ANN learns complex nonlinear relationships between employee characteristics and attrition by adjusting its internal weights during the training process. Compared to traditional machine learning algorithms, ANN is capable of capturing intricate relationships among multiple variables, making it suitable for employee attrition prediction.

6.5.1 ANN Architecture

Layer (type)	Output Shape	Param #
dense_1 (Dense)	(None, 8)	192
dense_2 (Dense)	(None, 16)	144
dense_3 (Dense)	(None, 2)	34
Total params: 370		
Trainable params: 370		
Non-trainable params: 0		

Figure 6.1: Artificial Neural Network Architecture Summary

Observation

The model summary shows three dense layers: the first with an output shape of 8 units and 192 trainable parameters, the second with 16 units and 144 parameters, and the output layer with 2 units and 34 parameters. The total trainable parameter count is 370, with no non-trainable parameters.

Business Interpretation

The architecture is intentionally lightweight — a small number of parameters relative to typical deep learning models. For a structured, tabular HR dataset of this size (1,470 records, 35 features), a compact network reduces the risk of overfitting while still being expressive enough to capture the non-linear relationships identified during exploratory analysis in Chapter 5.

6.5.2 Model Compilation

The ANN model was compiled using the following configuration:

Parameter	Value
Optimizer	Adam
Loss Function	Sparse Categorical Crossentropy
Evaluation Metric	Accuracy

The Adam optimizer was selected because of its efficient learning capability and faster convergence during training.

6.5.3 Model Training

The processed dataset was divided into training and testing subsets before training the Artificial Neural Network. During training, the model continuously updated its internal parameters to minimize prediction error and improve classification performance. The learning process was monitored using training accuracy and training loss across multiple epochs, as illustrated in Figures 6.2 and 6.3.

6.6 Model Performance

The trained Artificial Neural Network was evaluated using unseen testing data. The model achieved the following performance:

Metric	Result
Test Accuracy	88.45%
Test Loss	0.354

Discussion

The trained Artificial Neural Network achieved an overall prediction accuracy of 88.45%, indicating good predictive performance for employee attrition classification.

The model successfully learned relationships between employee demographics, compensation, workplace satisfaction, job involvement, work-life balance, and organizational characteristics.

Although the model demonstrated strong predictive capability, its performance may vary when applied to employee data from different organizations due to differences in workforce characteristics and organizational culture.

The model can therefore be considered a useful decision-support tool for Human Resource departments in identifying employees who may require proactive retention strategies.

6.6.1 Training Loss

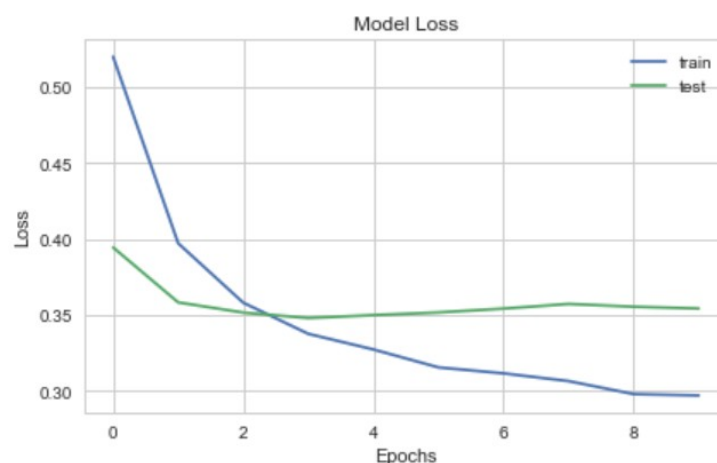


Figure 6.2: Training and Validation Loss

Observation

Training loss starts near 0.52 and declines steadily throughout training, reaching approximately 0.30 by the final epoch. Validation loss starts lower, near 0.40, drops quickly to around 0.35 by epoch 2, and then plateaus — fluctuating narrowly between roughly 0.35 and 0.36 for the remainder of training, with a slight uptick around epoch 7 before settling back down.

Interpretation

The two curves track closely for the first two to three epochs, after which training loss continues to fall while validation loss flattens out. This divergence is a mild early indicator of overfitting, though the gap remains small — validation loss never rises sharply, suggesting the model generalizes reasonably well within the scope of this dataset.

6.6.2 Training Accuracy

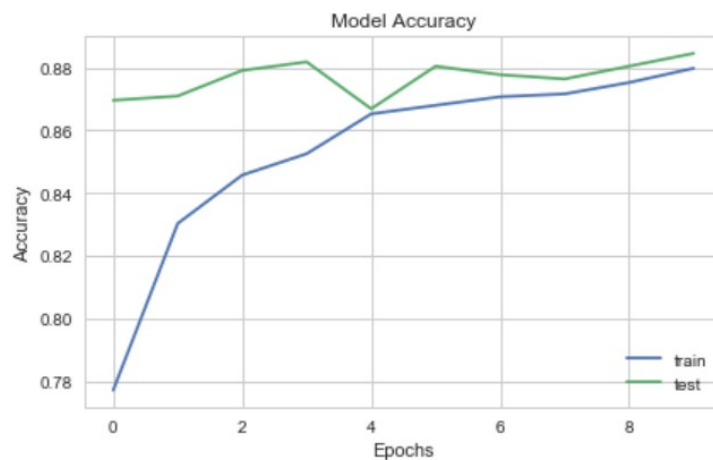


Figure 6.3: Training and Validation Accuracy

Observation

Training accuracy rises from approximately 0.778 at the first epoch to about 0.880 by the final epoch, in a smooth, steady climb. Validation accuracy starts higher, near 0.870, and moves in a narrower band between roughly 0.865 and 0.885, including a brief dip around epoch 4 before recovering and ending close to 0.885.

Interpretation

Validation accuracy remains consistently at or above training accuracy for most of the run, and the two converge by the final epochs. This is a favorable sign — it indicates the model is not simply memorizing the training set, and its performance on unseen data is stable rather than degrading over time.

6.7 Summary

The dataset was successfully prepared using feature selection, SMOTE, and feature scaling techniques before training an Artificial Neural Network. The trained model achieved an accuracy of 88.45%, indicating strong predictive capability for employee attrition. These results provide a solid analytical foundation for generating actionable business insights and supporting data-driven Human Resource decision-making.

CHAPTER 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

7.1 Introduction

The primary objective of Business Analytics is not only to analyze data but also to transform analytical findings into actionable business decisions. After performing exploratory data analysis and developing the Artificial Neural Network (ANN) model, several significant patterns influencing employee attrition were identified. These insights provide Human Resource (HR) departments with valuable information for improving employee retention, workforce planning, and organizational performance.

7.2 Key Business Insights

Insight 1: Employee Attrition is Relatively Low but Requires Attention

Observation

The majority of employees remained with the organization, while a smaller proportion experienced attrition.

Business Insight

Although the overall attrition rate is relatively low, organizations should focus on employees who exhibit characteristics associated with higher attrition risk. Early identification enables HR departments to implement proactive retention strategies before employees resign.

Insight 2: Monthly Income Influences Employee Retention

Observation

Employees who remained with the organization generally received higher monthly incomes compared to employees who left.

Business Insight

Compensation plays a significant role in employee retention. Employees receiving lower salaries may perceive limited financial growth opportunities and become more likely to seek alternative employment.

Insight 3: Younger Employees Demonstrate Higher Attrition

Observation

Employees who experienced attrition generally belonged to relatively younger age groups.

Business Insight

Early-career employees are more likely to explore alternative employment opportunities. Organizations should strengthen onboarding programs, career development initiatives, and mentorship opportunities to improve long-term retention.

Insight 4: Job Level Influences Employee Attrition

Observation

Employees at Job Level 1 recorded the highest attrition rate (26.34%), whereas employees occupying higher job levels exhibited significantly lower attrition.

Business Insight

Career progression and professional growth contribute substantially to employee retention. Organizations should establish transparent promotion policies and clearly communicate career advancement opportunities.

Insight 5: Work-Life Balance Impacts Employee Retention***Observation***

Employees reporting poor work-life balance (rating 1) experienced noticeably higher attrition, at 31.25%.

Business Insight

Maintaining a healthy balance between professional responsibilities and personal well-being contributes significantly to employee satisfaction and long-term organizational commitment.

Insight 6: Job Involvement Influences Employee Commitment***Observation***

Employees with lower job involvement exhibited considerably higher attrition (33.73%) than employees actively engaged in their work (9.03% at the highest involvement rating).

Business Insight

Organizations should encourage employee participation, recognize individual contributions, and create opportunities for professional involvement to strengthen employee commitment.

Insight 7: Workplace Environment Affects Employee Satisfaction***Observation***

Employees reporting lower environment satisfaction demonstrated higher attrition rates, decreasing steadily from 25.35% at the lowest rating to 13.45% at the highest.

Business Insight

A positive workplace environment promotes employee engagement, improves morale, and contributes to long-term retention.

Insight 8: Overtime May Increase Attrition Risk***Observation***

Approximately 416 employees reported working overtime, representing close to a quarter of the workforce.

Business Insight

While overtime may occasionally be necessary, prolonged excessive workloads can contribute to stress, burnout, and employee dissatisfaction. HR departments should regularly monitor overtime patterns and ensure appropriate workload distribution.

7.3 Business Recommendations

Based on the analysis conducted in this project, the following recommendations are proposed to improve employee retention:

- Implement structured career development programs for employees in entry-level positions to improve long-term organizational commitment.
- Conduct periodic salary benchmarking and performance-based compensation reviews to ensure competitive remuneration.
- Strengthen employee engagement initiatives through regular feedback sessions, recognition programs, and professional development opportunities.
- Promote healthy work-life balance by monitoring overtime, encouraging flexible work practices where feasible, and supporting employee well-being initiatives.
- Conduct regular employee satisfaction surveys to identify workplace concerns before they result in employee turnover.
- Develop mentoring and training programs for newly recruited employees to improve organizational integration and career confidence.
- Utilize predictive analytics models to identify employees at higher attrition risk and initiate timely retention interventions.

7.4 Business Value of the Project

The developed predictive model demonstrates how Business Analytics can support strategic Human Resource decision-making. The outcomes of this project enable organizations to:

- Identify employees at higher risk of attrition.
- Reduce recruitment and training costs.
- Improve employee satisfaction.
- Support workforce planning.
- Enhance organizational productivity.
- Promote data-driven HR decision-making.
- Strengthen long-term employee retention strategies.

Chapter Summary

The business insights generated through this analysis demonstrate that employee attrition is influenced by multiple organizational and demographic factors, including compensation, job level, work-life balance, job involvement, and workplace satisfaction. These findings provide Human Resource departments with practical guidance for improving employee retention and creating a more engaged and productive workforce.

CHAPTER 8: CONCLUSION, LIMITATIONS, FUTURE SCOPE AND REFERENCES

8.1 Conclusion

Employee attrition is a critical organizational challenge that affects workforce stability, operational efficiency, recruitment costs, and overall business performance. Identifying employees who are at risk of leaving enables organizations to implement proactive retention strategies and improve workforce planning.

In this project, the IBM HR Analytics Employee Attrition & Performance dataset was analyzed using Business Analytics techniques and Machine Learning methodologies. The project involved data preprocessing, exploratory data analysis, feature engineering, handling class imbalance using SMOTE, feature scaling, and the development of an Artificial Neural Network model.

The developed ANN model achieved an overall prediction accuracy of 88.45%, demonstrating its effectiveness in predicting employee attrition. The analysis revealed that factors such as monthly income, overtime, job level, work-life balance, job involvement, environment satisfaction, relationship satisfaction, and employee experience significantly influence attrition.

The insights obtained from this analysis can assist Human Resource departments in making informed decisions, improving employee engagement, and implementing effective retention strategies. Overall, the project demonstrates how Business Analytics and Machine Learning can support evidence-based Human Resource decision-making and contribute to organizational success.

8.2 Limitations

- The analysis is based on the IBM HR Analytics Employee Attrition dataset, which may not fully represent the workforce characteristics of every organization.
- The dataset provides a static snapshot of employee information and does not capture changes over time.
- External factors such as organizational culture, economic conditions, leadership style, employee health, and personal circumstances were not included in the dataset.
- The predictive model was developed using historical employee data and may require retraining when applied to new organizational environments.
- The project focused on structured employee data and did not include qualitative information such as employee feedback or exit interview responses.

8.3 Future Scope

- Develop interactive HR dashboards using Power BI or Tableau for real-time employee monitoring.
- Compare the ANN model with advanced machine learning algorithms such as XGBoost, LightGBM, and CatBoost.
- Integrate historical employee records to forecast attrition trends over the next 5–10 years.
- Develop a web-based HR decision-support system that predicts employee attrition in real time.
- Incorporate employee feedback, engagement surveys, and performance reviews to improve prediction accuracy.

- Apply Explainable AI (XAI) techniques to improve transparency in machine learning predictions and support HR decision-making.

8.4 References

- IBM HR Analytics Employee Attrition & Performance Dataset (Kaggle)
- TensorFlow Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Seaborn Documentation
- Scikit-learn paper
- Deep Learning with Python